

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 October 2023 Worked Solutions

October 2023 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com

P2

PAST PAPER

WMA12/01 October 2023

October 2023 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Proof

1. Given that a , b and c are integers greater than 0 such that

- $c = 3a + 1$
- $a + b + c = 15$

prove, by exhaustion, that the product abc is always a multiple of 4

You may use the table below to illustrate your answer.

(3)

You may not need to use all rows of this table.

a	b	c	abc

(Total for Question 1 is 3 marks)

(Total for Question 1 is 3 marks)

Worked Solution - Question 1

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: The required statement has been proved.

Final answer

The required statement has been proved.

Question 2

Sequences & Series

2. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 3$$

$$u_{n+1} = 2 - \frac{4}{u_n}$$

- (a) Find the value of u_2 , the value of u_3 and the value of u_4

(3)

- (b) Find the value of

$$\sum_{r=1}^{100} u_r$$

(2)

(Total for Question 2 is 5 marks)

(Total for Question 2 is 5 marks)

Worked Solution - Question 2

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: = - = (b) 100 1 2 33 3 4 3
 $3 r r u = = \times + + - + = 8$

Final answer

= - = (b) 100 1 2 33 3 4 3 3 r r u = = \times + + - + = 8

Question 3

Trigonometric Equations

3. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (a) Solve, for $0 < \theta \leq 360^\circ$ the equation

$$2 \tan \theta + 3 \sin \theta = 0$$

giving your answers, as appropriate, to one decimal place.

(5)

- (b) Hence, or otherwise, find the smallest positive solution of

$$2 \tan (2x + 40^\circ) + 3 \sin (2x + 40^\circ) = 0$$

giving your answer to one decimal place.

(2)

(Total for Question 3 is 7 marks)

(Total for Question 3 is 7 marks)

Worked Solution - Question 3

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 4

Polynomials

4.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + ax^2 - 29x + b$$

where a and b are constants.

Given that $(2x + 1)$ is a factor of $f(x)$,

(a) show that

$$a + 4b = -56 \quad (2)$$

Given also that when $f(x)$ is divided by $(x - 2)$ the remainder is -25

(b) find a second simplified equation linking a and b . (2)

(c) Hence, using algebra and showing your working,

(i) find the value of a and the value of b ,

(ii) fully factorise $f(x)$. (5)

(Total for Question 4 is 9 marks)

(Total for Question 4 is 9 marks)

Worked Solution - Question 4

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: + - + cso

Final answer

+ - + CSO

Question 5

Laws of Logarithms

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve

$$3^a = 70$$

giving the answer to 3 decimal places.

(2)

(ii) Find the exact value of b such that

$$4 + 3 \log_3 b = \log_3 5b$$

(4)

(Total for Question 5 is 6 marks)

(Total for Question 5 is 6 marks)

Worked Solution - Question 5

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: $5 \log b =$

Final answer

 $5 \log b =$

Question 6

Integration

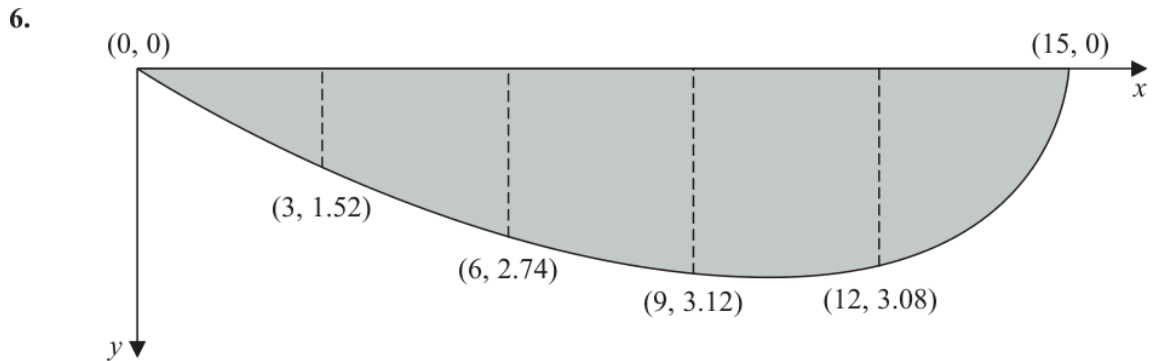


Figure 1

A river is being studied.

At one particular place, the river is 15 m wide.

The depth, y metres, of the river is measured at a point x metres from one side of the river.

Figure 1 shows a plot of the cross-section of the river and the coordinate values (x, y)

- (a) Use the trapezium rule with all the y values given in Figure 1 to estimate the cross-sectional area of the river.

(3)

The water in the river is modelled as flowing at a constant speed of 1.5 m s^{-1} across the whole of the cross-section.

- (b) Use the model and the answer to part (a) to estimate the volume of water flowing through this section of the river each minute, giving your answer in m^3 to 2 significant figures.

(2)

Assuming the model,

- (c) state, giving a reason for your answer, whether your answer for part (b) is an overestimate or an underestimate of the true volume of water flowing through this section of the river each minute.

(1)

(Total for Question 6 is 6 marks)

(Total for Question 6 is 6 marks)

Worked Solution - Question 6

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 7

7.

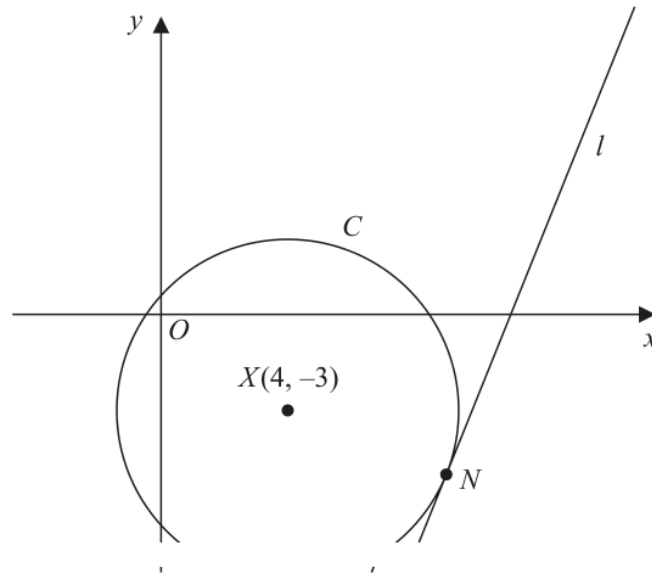


Figure 2

Figure 2 shows a sketch of

- the circle C with centre $X(4, -3)$
- the line l with equation $y = \frac{5}{2}x - \frac{55}{2}$

Given that l is the tangent to C at the point N ,

(a) show that an equation for the straight line passing through X and N is

$$2x + 5y + 7 = 0$$

(3)

(b) Hence find

- the coordinates of N ,
- an equation for C .

(5)

(Total for Question 7 is 8 marks)

(Total for Question 7 is 8 marks)

Worked Solution - Question 7

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: $- + - + () () 2 2 4 3 29 x y$
 $- + + =$

Final answer

$- + - + () () 2 2 4 3 29 x y - + + =$

Question 8

Arithmetic Sequences

8. In a large theatre there are n rows of seats, where n is a constant.

The number of seats in the first row is a , where a is a constant.

In each subsequent row there are 4 more seats than in the previous row so that

- in the 2nd row there are $(a + 4)$ seats
- in the 3rd row there are $(a + 8)$ seats
- the number of seats in each row form an **arithmetic** sequence

Given that the **total** number of seats in the first 10 rows is 360

- (a) find the value of a .

(2)

Given also that the total number of seats in the n rows is 2146

- (b) show that

$$n^2 + 8n - 1073 = 0$$

(2)

- (c) Hence

- (i) state the number of rows of seats in the theatre,
- (ii) find the maximum number of seats in any one row.

(3)

(Total for Question 8 is 7 marks)

(Total for Question 8 is 7 marks)

Worked Solution - Question 8

1. Use the arithmetic formula

Identify the first term, common difference, and required term or sum from the question.

2. Substitute and solve

Use the arithmetic term or sum formula, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 9

Applications of Differentiation

9.

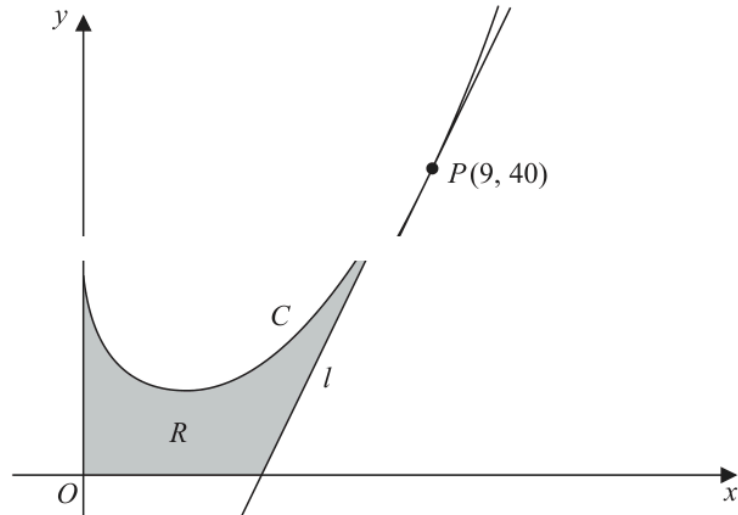


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve C with equation

$$y = \frac{2}{3}x^2 - 9\sqrt{x} + 13 \quad x \geq 0$$

(a) Find, using calculus, the range of values of x for which y is increasing.

(4)

The point P lies on C and has coordinates $(9, 40)$.

The line l is the tangent to C at the point P .

The finite region R , shown shaded in Figure 3, is bounded by the curve C , the line l , the x -axis and the y -axis.

(b) Find, using calculus, the exact area of R .

(8)

(Total for Question 9 is 12 marks)

(Total for Question 9 is 12 marks)

Worked Solution - Question 9

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: 1600 857 117 21 21 - =

Final answer

1600 857 117 21 21 - =

Question 10

Binomial Expansion

10. (i) (a) Find, in ascending powers of x , the 2nd, 3rd and 5th terms of the binomial expansion of

$$(3 + 2x)^6 \quad (3)$$

For a particular value of x , these three terms form consecutive terms in a geometric series.

- (b) Find this value of x . (3)

- (ii) In a **different** geometric series,

- the first term is $\sin^2 \theta$
- the common ratio is $2 \cos \theta$
- the sum to infinity is $\frac{8}{5}$

- (a) Show that

$$5 \cos^2 \theta - 16 \cos \theta + 3 = 0 \quad (3)$$

- (b) Hence find the exact value of the 2nd term in the series. (3)

(Total for Question 10 is 12 marks)

(Total for Question 10 is 12 marks)

Worked Solution - Question 10

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: $\times = - \times \times \theta \theta 48 125 =$

Final answer

$\times = - \times \times \theta \theta 48 125 =$